

than a microwave radar as the former provides better collimation, which makes high angular resolution possible. Having the advantage of greater radiant brightness and the fact that it is directional even after travelling long distances, the size of emitting system is gradually reduced. A laser range-finder of medium range (up to 10km) is used in several defence areas, including:

1. Tank laser rangefinders for battle tanks
2. Portable laser rangefinders used in field artillery fire-control systems
3. Air-borne laser rangefinders for air forces
4. Laser walkie-talkie rangefinders

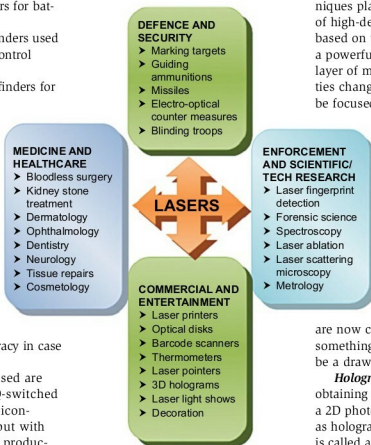
Another application of laser is lidars (or, laser radars). These are better than microwaves, as lasers can be focused with lenses and mirrors easily while microwaves need huge antennae and components. Moreover, dimension and distance of target can be obtained with high accuracy in case of lidars.

The types of lasers used are carbon-dioxide lasers, Q-switched or gallium-arsenide semiconductor. High power output with requisite spectral purity production capacity of CO₂ laser is better for this purpose. High frequency of CO₂ lasers also produces high Doppler shift even from slow-moving targets. Fine beam width and high Doppler shift give CO₂ lasers an unparalleled imaging capability. Radar systems are used for measuring radial velocities to track low-flying aircraft and slow-moving objects.

Then, there are laser-guided anti-missile systems that can be guided by an infrared beam emitted

from a laser, with extremely small divergence that can be achieved in the following four ways:

1. The laser beam is used to illuminate the target tank. The anti-missile system then homes in on the target, as the latter becomes a source of back-scattered radiation.
2. A laser beam is used to provide guidance instructions to the missile.



Applications of lasers

3. The missile itself carries a laser scanner and a seeker for active homing on target.
4. The missile rides the laser beam towards the target.

In an anti-missile defence setup, a laser is used to dispose of the energy of the warhead, by partially damaging the missile. Tremendous energy is required to completely destroy the missile. According to predictions, lasers will ultimately

make inter-continental ballistic missiles obsolete.

There are many limitations, however, to the laser playing an anti-missile role. Huge power stations are required to produce huge power lasers. CO₂ and chemical lasers developed in the USA and Russia produce huge power in continuous mode, which is sufficient to destroy enemy battle tank.

Data storage. Optical techniques play a major role in storage of high-density data. These are based on the principle that, when a powerful laser illuminates a thin layer of metal, its optical properties change. As a laser beam can be focused on points smaller than one micro diameter, it takes less than one square micro to record one bit of data, that is, 100 million per square cm. There are laser CDs and DVDs available in the market with the required data stored in the form of audio, video, document, etc. Interestingly, erasable disks

are now coming into the picture, something that was considered to be a drawback.

Holography. The technique of obtaining an image of a 3D body on a 2D photographic plate is termed as holography. The stored image is called a hologram. The whole process is based on the interference produced by the interaction of two-beam monochromatic light waves under certain conditions.

The hologram is recorded when a part of the emitted laser beam or reference beam goes directly to the photographic plate and the other part being object beam is reflected from the object and made to fall on the photographic plate, hence interfering with the reference beam to produce an interference pattern, which is called a